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ERG Analysis API v2.4.0

Tier 3 — T3-B: Normative Reference Traceability Matrix

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Project	ERG Analysis API — AI Fairness CIO (Charity Reg. No. 1218464)
Pipeline version	v2.4.0 (PIPELINE_VERSION = "2.4.0")
Normative JSON	data/normative_data_baker2025.json (schema_version 1.0)
Primary reference	Baker RA, et al. (2025). Doc Ophthalmol 150(2):47–64. DOI: 10.1007/s10633-025-10009-2. N=407.
Age strata	le35 (≤35y) 36to59 (36–59y) ge60 (≥60y)
Electrodes	Gold Foil (primary norms) DTL Fiber (Deming-corrected to Gold Foil equivalent)
OSF pre-registration	https://doi.org/10.17605/OSF.IO/6WA42
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Status	Sign-off by Hamid Tavakoli

1. Purpose

This document provides a complete audit trail from the Baker et al. (2025) published normative data through every transformation step to the final Z-score and traffic-light output on the clinical report. For every normative value in the JSON file, the cross-walk records the exact JSON key path, the numerical value (μ and σ), the pipeline variable that consumes it, the report field where it appears, and a traceability status.

Three normative data structures are covered: (1) protocol normative ranges for five ISCEV protocols, two electrode types, and three age strata (Section 4); (2) DTL Deming regression transforms from Baker 2025 Table 2 (Section 5); and (3) the b/a ratio reference for DA 3 (Section 6).

2. End-to-End Data Flow

Normative data flows through the pipeline in four stages:

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- Stage 1 — Load. `_find_norm_json()` locates `normative_data_baker2025.json`; `json.load()` reads it; `schema_version` assertion confirms v1.0; `_to_tuple()` converts {mu, sigma} dicts to tuples. `electrode_transform` block loaded separately.
- Stage 2 — Select. `compute_z_scores()` calls `_age_stratum(age_years)` to resolve stratum key, then `_get_ref(electrode, protocol, stratum)` to retrieve the (mu, sigma) tuple.
- Stage 3 — Transform (DTL only). `_apply_dtl_transform()` inverts Deming regression: $GFE_{equivalent} = (STE_{measured} - intercept) / slope$. Applied to amplitude parameters only.
- Stage 4 — Score and classify. $Z = (value - \mu) / \sigma$. `generate_traffic_light()` maps $\max|Z|$ to Green / Amber / Red. Output in Layer 2 summary, Layer 4 audit, FHIR, and PDF.

3. Findings Summary

Result	Count
PASS	5
AMBER	3 → 0 after remediation
FAIL	0
TOTAL	8

All three AMBER findings from the initial audit (F1, F4, F6) were remediated during Tier 3. No FAIL findings were raised. All 8 findings are now PASS.

- F1 — Schema version assertion added at JSON load time (`assert schema_version == "1.0"`).
- F4 — LA 3 a_amp DTL borderline $r^2=0.68$ now flagged in Layer 4 audit (`_la3_a_amp_dtl_r2_borderline = True`).
- F6 — b/a ratio constants now loaded from JSON `ba_ratio` block; no hardcoding remains.

4. Protocol Normative Reference Cross-walk

Each row maps one (protocol, stratum, parameter) cell from the normative JSON to the pipeline variable and report field. Grey rows indicate null entries — parameters not defined for that protocol per ISCEV 2022 (e.g., a-wave for DA 0.01). DTL Fiber shares Gold Foil normative values; the Deming correction is applied at Z-score time and is documented in Section 5.

Note: 36 to 59 stratum is labelled '36–59y' in the report and stored as key '36to59' in the JSON. The pipeline maps age 36–59 → '36to59'.

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Protocol	Stratum	Param	Unit	μ	σ	JSON key path	Pipeline variable	Report field	Status
Protocol: DA 0.01 — Gold Foil and DTL Fiber (DTL shares Gold Foil norms; Deming correction applied at Z-score time)									
DA 0.01	≤35y	a_amp	μV	null	null	protocols["DA 0.01"]["gold_foil"]["le35"]["a_amp"]	— not used	— not rendered	NA
		a_imp	ms	null	null	protocols["DA 0.01"]["gold_foil"]["le35"]["a_imp"]	— not used	— not rendered	NA
		b_amp	μV	203.85	40.99	protocols["DA 0.01"]["gold_foil"]["le35"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	84.65	6.17	protocols["DA 0.01"]["gold_foil"]["le35"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
DA 0.01	36–59y	a_amp	μV	null	null	protocols["DA 0.01"]["gold_foil"]["36to59"]["a_amp"]	— not used	— not rendered	NA
		a_imp	ms	null	null	protocols["DA 0.01"]["gold_foil"]["36to59"]["a_imp"]	— not used	— not rendered	NA
		b_amp	μV	189.48	38.42	protocols["DA 0.01"]["gold_foil"]["36to59"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	90.33	7.6	protocols["DA 0.01"]["gold_foil"]["36to59"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
DA 0.01	≥60y	a_amp	μV	null	null	protocols["DA 0.01"]["gold_foil"]["ge60"]["a_amp"]	— not used	— not rendered	NA
		a_imp	ms	null	null	protocols["DA 0.01"]["gold_foil"]["ge60"]["a_imp"]	— not used	— not rendered	NA
		b_amp	μV	185.53	46.36	protocols["DA 0.01"]["gold_foil"]["ge60"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	97.29	7.81	protocols["DA 0.01"]["gold_foil"]["ge60"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
Protocol: DA 3 — Gold Foil and DTL Fiber (DTL shares Gold Foil norms; Deming correction applied at Z-score time)									
DA 3	≤35y	a_amp	μV	175.7	30.93	protocols["DA 3"]["gold_foil"]["le35"]["a_amp"]	features["a_wave_amplitude_uv"] abs() applied DTL: Deming inversion	Layer 2 metrics table Row 1: A-Wave Amplitude	PASS
		a_imp	ms	14.98	0.65	protocols["DA 3"]["gold_foil"]["le35"]["a_imp"]	features["a_wave_implicit_time_ms"] No transform	Layer 4 audit JSON z_scores["a_wave_implicit_time"]	PASS
		b_amp	μV	281.45	53.29	protocols["DA 3"]["gold_foil"]["le35"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	51.41	3.18	protocols["DA 3"]["gold_foil"]["le35"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS

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Protocol	Stratum	Param	Unit	μ	σ	JSON key path	Pipeline variable	Report field	Status
DA 3	36–59y	a_amp	μ V	157.86	31.63	protocols["DA 3"]["gold_foil"]["36to59"]["a_amp"]	features["a_wave_amplitude_uv"] abs() applied DTL: Deming inversion	Layer 2 metrics table Row 1: A-Wave Amplitude	PASS
		a_imp	ms	15.69	0.68	protocols["DA 3"]["gold_foil"]["36to59"]["a_imp"]	features["a_wave_implicit_time_ms"] No transform	Layer 4 audit JSON z_scores["a_wave_implicit_time"]	PASS
		b_amp	μ V	272	51.91	protocols["DA 3"]["gold_foil"]["36to59"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	51.45	2.9	protocols["DA 3"]["gold_foil"]["36to59"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
DA 3	≥60y	a_amp	μ V	141.46	35.25	protocols["DA 3"]["gold_foil"]["ge60"]["a_amp"]	features["a_wave_amplitude_uv"] abs() applied DTL: Deming inversion	Layer 2 metrics table Row 1: A-Wave Amplitude	PASS
		a_imp	ms	16.08	0.82	protocols["DA 3"]["gold_foil"]["ge60"]["a_imp"]	features["a_wave_implicit_time_ms"] No transform	Layer 4 audit JSON z_scores["a_wave_implicit_time"]	PASS
		b_amp	μ V	259.25	62.43	protocols["DA 3"]["gold_foil"]["ge60"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	52.45	3.02	protocols["DA 3"]["gold_foil"]["ge60"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
Protocol: DA 10 — Gold Foil and DTL Fiber (DTL shares Gold Foil norms; Deming correction applied at Z-score time)									
DA 10	≤35y	a_amp	μ V	202.99	34.59	protocols["DA 10"]["gold_foil"]["le35"]["a_amp"]	features["a_wave_amplitude_uv"] abs() applied DTL: Deming inversion	Layer 2 metrics table Row 1: A-Wave Amplitude	PASS
		a_imp	ms	11.99	0.98	protocols["DA 10"]["gold_foil"]["le35"]["a_imp"]	features["a_wave_implicit_time_ms"] No transform	Layer 4 audit JSON z_scores["a_wave_implicit_time"]	PASS
		b_amp	μ V	291.01	50.92	protocols["DA 10"]["gold_foil"]["le35"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	51.82	3.08	protocols["DA 10"]["gold_foil"]["le35"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
DA 10	36–59y	a_amp	μ V	188.6	35.09	protocols["DA 10"]["gold_foil"]["36to59"]["a_amp"]	features["a_wave_amplitude_uv"] abs() applied DTL: Deming inversion	Layer 2 metrics table Row 1: A-Wave Amplitude	PASS
		a_imp	ms	12.83	1.07	protocols["DA 10"]["gold_foil"]["36to59"]["a_imp"]	features["a_wave_implicit_time_ms"] No transform	Layer 4 audit JSON z_scores["a_wave_implicit_time"]	PASS
		b_amp	μ V	283.69	50.41	protocols["DA 10"]["gold_foil"]["36to59"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS

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Protocol	Stratum	Param	Unit	μ	σ	JSON key path	Pipeline variable	Report field	Status
		b_imp	ms	51.94	3.04	protocols["DA 10"]["gold_foil"]["36to59"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
DA 10	≥60y	a_amp	μV	170.81	39.57	protocols["DA 10"]["gold_foil"]["ge60"]["a_amp"]	features["a_wave_amplitude_uv"] abs() applied DTL: Deming inversion	Layer 2 metrics table Row 1: A-Wave Amplitude	PASS
		a_imp	ms	13.42	1.09	protocols["DA 10"]["gold_foil"]["ge60"]["a_imp"]	features["a_wave_implicit_time_ms"] No transform	Layer 4 audit JSON z_scores["a_wave_implicit_time"]	PASS
		b_amp	μV	275.75	62.12	protocols["DA 10"]["gold_foil"]["ge60"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	52.21	3.32	protocols["DA 10"]["gold_foil"]["ge60"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
Protocol: LA 3 — Gold Foil and DTL Fiber (DTL shares Gold Foil norms; Deming correction applied at Z-score time)									
LA 3	≤35y	a_amp	μV	26.39	6.73	protocols["LA 3"]["gold_foil"]["le35"]["a_amp"]	features["a_wave_amplitude_uv"] abs() applied DTL: Deming inversion	Layer 2 metrics table Row 1: A-Wave Amplitude	PASS
		a_imp	ms	14.03	0.53	protocols["LA 3"]["gold_foil"]["le35"]["a_imp"]	features["a_wave_implicit_time_ms"] No transform	Layer 4 audit JSON z_scores["a_wave_implicit_time"]	PASS
		b_amp	μV	124.93	36.06	protocols["LA 3"]["gold_foil"]["le35"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	28.77	1	protocols["LA 3"]["gold_foil"]["le35"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
LA 3	36–59y	a_amp	μV	23.81	6.35	protocols["LA 3"]["gold_foil"]["36to59"]["a_amp"]	features["a_wave_amplitude_uv"] abs() applied DTL: Deming inversion	Layer 2 metrics table Row 1: A-Wave Amplitude	PASS
		a_imp	ms	14.3	0.63	protocols["LA 3"]["gold_foil"]["36to59"]["a_imp"]	features["a_wave_implicit_time_ms"] No transform	Layer 4 audit JSON z_scores["a_wave_implicit_time"]	PASS
		b_amp	μV	108.9	27.32	protocols["LA 3"]["gold_foil"]["36to59"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	29.05	1.1	protocols["LA 3"]["gold_foil"]["36to59"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
LA 3	≥60y	a_amp	μV	22.33	5.82	protocols["LA 3"]["gold_foil"]["ge60"]["a_amp"]	features["a_wave_amplitude_uv"] abs() applied DTL: Deming inversion	Layer 2 metrics table Row 1: A-Wave Amplitude	PASS
		a_imp	ms	14.4	0.64	protocols["LA 3"]["gold_foil"]["ge60"]["a_imp"]	features["a_wave_implicit_time_ms"] No transform	Layer 4 audit JSON z_scores["a_wave_implicit_time"]	PASS

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Protocol	Stratum	Param	Unit	μ	σ	JSON key path	Pipeline variable	Report field	Status
		b_amp	μV	96.31	27.73	protocols["LA 30 Hz"]["gold_foil"]["ge60"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	29.55	1.26	protocols["LA 30 Hz"]["gold_foil"]["ge60"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
Protocol: LA 30 Hz — Gold Foil and DTL Fiber (DTL shares Gold Foil norms; Deming correction applied at Z-score time)									
LA 30 Hz	≤35y	a_amp	μV	null	null	protocols["LA 30 Hz"]["gold_foil"]["le35"]["a_amp"]	— not used	— not rendered	NA
		a_imp	ms	null	null	protocols["LA 30 Hz"]["gold_foil"]["le35"]["a_imp"]	— not used	— not rendered	NA
		b_amp	μV	92.6	29.11	protocols["LA 30 Hz"]["gold_foil"]["le35"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	25.5	0.89	protocols["LA 30 Hz"]["gold_foil"]["le35"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
LA 30 Hz	36–59y	a_amp	μV	null	null	protocols["LA 30 Hz"]["gold_foil"]["36to59"]["a_amp"]	— not used	— not rendered	NA
		a_imp	ms	null	null	protocols["LA 30 Hz"]["gold_foil"]["36to59"]["a_imp"]	— not used	— not rendered	NA
		b_amp	μV	80.72	22.96	protocols["LA 30 Hz"]["gold_foil"]["36to59"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	26	1.28	protocols["LA 30 Hz"]["gold_foil"]["36to59"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS
LA 30 Hz	≥60y	a_amp	μV	null	null	protocols["LA 30 Hz"]["gold_foil"]["ge60"]["a_amp"]	— not used	— not rendered	NA
		a_imp	ms	null	null	protocols["LA 30 Hz"]["gold_foil"]["ge60"]["a_imp"]	— not used	— not rendered	NA
		b_amp	μV	71.13	20.64	protocols["LA 30 Hz"]["gold_foil"]["ge60"]["b_amp"]	features["b_wave_amplitude_uv"] DTL: Deming inversion	Layer 2 metrics table Row 2: B-Wave Amplitude	PASS
		b_imp	ms	26.45	1.48	protocols["LA 30 Hz"]["gold_foil"]["ge60"]["b_imp"]	features["b_wave_implicit_time_ms"] No transform	Layer 2 metrics table Row 3: B-Wave Implicit Time	PASS

5. DTL Deming Regression Transform Cross-walk

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Baker et al. (2025) Table 2 provides Deming regression coefficients for converting DTL Fiber amplitude measurements to Gold Foil equivalent. The pipeline inverts this: $GFE = (STE - \text{intercept}) / \text{slope}$, then Z-scores against Gold Foil norms. Peak time parameters (a_imp, b_imp) are not transformed (bias ≤ 1.6 ms per Baker 2025).

Protocol	Param	Slope	Intercept	r^2	n	Method	Inversion formula: $GFE = (STE - \text{intercept}) / \text{slope}$	Status
Parameter: a_amp — amplitude parameters only; peak times (a_imp, b_imp) are not transformed								
DA 0.01	—	—	—	—	—	—	—	NA
DA 3	a_amp	0.56	8.24	0.79	49	Deming	$GFE = (STE - 8.24) / 0.56$	PASS
DA 10	a_amp	0.55	9.93	0.74	50	Deming	$GFE = (STE - 9.93) / 0.55$	PASS
LA 3	a_amp	0.55	1.15	0.68	52	Deming	$GFE = (STE - 1.15) / 0.55$ ⚠ Borderline $r^2=0.68$ — Layer 4 flag _la3_a_amp_dtl_r2_borderline set	AMBER
LA 30 Hz	—	—	—	—	—	—	—	NA
Parameter: b_amp — amplitude parameters only; peak times (a_imp, b_imp) are not transformed								
DA 0.01	b_amp	0.46	24.38	0.73	46	Deming	$GFE = (STE - 24.38) / 0.46$	PASS
DA 3	b_amp	0.48	36.75	0.73	47	Deming	$GFE = (STE - 36.75) / 0.48$	PASS
DA 10	b_amp	0.48	37.92	0.72	49	Deming	$GFE = (STE - 37.92) / 0.48$	PASS
LA 3	b_amp	0.65	-3	0.81	52	Deming	$GFE = (STE - -3) / 0.65$	PASS
LA 30 Hz	b_amp	0.73	-9.54	0.85	52	Deming	$GFE = (STE - -9.54) / 0.73$	PASS

6. b/a Ratio Reference Cross-walk

The b/a ratio reference is a top-level ba_ratio object in the normative JSON, separate from the protocols block. Applies to DA 3 only. Normal range 1.8–3.5 modelled as Gaussian (mean=2.65, SD=0.425). Following T3-B F6 remediation, these constants are loaded from the JSON at instantiation rather than hardcoded.

Item	Value / Reference	Status	Notes
JSON key path	ba_ratio (top-level object in normative_data_baker2025.json)	PASS	

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Item	Value / Reference	Status	Notes
mean	ba_ratio.mean = 2.65	PASS	
sigma	ba_ratio.sigma = 0.425	PASS	
Normal range	1.8–3.5 (mean $\pm$ 2SD: 2.65 $\pm$ 0.850)	PASS	
Protocol scope	DA 3 only (ba_ratio.protocol = "DA 3")	PASS	
Pipeline variable	features["ba_ratio"] = b_wave_raw / abs(a_wave_amplitude) computed in ERGFeatureExtractor.extract_features()	PASS	
JSON load (T3-B F6 patch)	self.BA_RATIO_MEAN = _norm["ba_ratio"]["mean"] self.BA_RATIO_SD = _norm["ba_ratio"]["sigma"] Loaded at ERGReportGenerator instantiation — no hardcoding	PASS	Patched in T3-B remediation: b/a ratio constants now loaded from JSON, not hardcoded
Z-score computation	compute_z_scores() line ~1312 BA_RATIO_MEAN and BA_RATIO_SD from JSON $Z = (ba_ratio - mean) / sigma$	PASS	
Report field	Layer 2 metrics table Row 4: B/A Ratio erg_report_generator.py param_defs	PASS	
DTL transform	NOT applied b/a ratio is dimensionless; amplitude transforms cancel	PASS	

7. Traceability Findings

#	Finding	Evidence	Status	Notes
F1	Schema version assertion	JSON schema_version = "1.0". Pipeline loader now asserts schema_version == "1.0" at load time (T3-B F1 patch). Raises AssertionError with clear message if future incompatible schema is loaded.	PASS	F1 patch applied: assert _norm.get("schema_version") == "1.0" added to ERGReportGenerator.__init__()

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#	Finding	Evidence	Status	Notes
F2	Electrode note — DTL shares Gold Foil norms	electrode_note field documents DTL uses Gold Foil norms with Deming correction. Pipeline confirms: ZSCORE_SUPPORTED_ELECTRODES includes dtl_fiber; _apply_dtl_transform() inverts Baker 2025 Table 2.	PASS	Traceability complete. Note field provides audit trail for the design decision.
F3	Null convention honoured	null_convention documented. _to_tuple() returns None for null values. compute_z_scores() guards with ref.get(param) is not None before computing Z.	PASS	Null handling is defensive and consistent.
F4	LA 3 a_amp Deming $r^2 = 0.68$ — borderline	Borderline r^2 documented in JSON note field. Pipeline now sets z_scores["la3_a_amp_dtl_r2_borderline"] = True in Layer 4 audit when this transform is applied (T3-B F4 patch).	PASS	F4 patch applied: flag set in compute_z_scores() for LA 3 a_amp on DTL electrode.
F5	Peak times: no Deming transform	electrode_transform.applies_to confirms amplitude only. a_imp and b_imp transform entries are null. _apply_dtl_transform() returns raw value when param_transform is None.	PASS	Correct per Baker 2025: peak time bias ≤ 1.6 ms; no transform needed.
F6	b/a ratio loaded from JSON	ba_ratio.mean and ba_ratio.sigma now loaded from normative JSON at instantiation (T3-B F6 patch). JSON is the single source of truth for all normative values.	PASS	F6 patch applied: self.BA_RATIO_MEAN = _norm["ba_ratio"]["mean"]; self.BA_RATIO_SD = _norm["ba_ratio"]["sigma"]
F7	N=407 and DOI in JSON metadata	n_subjects = 407; doi = "10.1007/s10633-025-10009-2". Written to Layer 3 normative_meta. Displayed in PDF subtitle.	PASS	Full traceability from report to primary source.
F8	Age strata keys match pipeline mapping	JSON age_strata = ["le35", "36to59", "ge60"]. Pipeline _age_stratum() maps correctly. Key mapping exact; no mismatch.	PASS	Verified across all 41 Tier 1 synthetic cases.

8. Sign-off

All three AMBER findings (F1, F4, F6) were remediated during Tier 3. The document is classified DRAFT pending Hamid Tavakoli counter-signature.

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Role	Name	Date
T3-B Auditor (H.T)	AI Fairness CIO ERG Pipeline	24 June 2026
Founder & CEO (Approval)	Hamid Tavakoli	26 June 2026

End of T3-B Normative Reference Traceability Matrix v1.0 — AI Fairness CIO (Reg. No. 1218464)